EEECE7352 Computer Architecture

Homework 4

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Part B: (15 Points)

For the next part of this assignment, using the trace provided as input to DineroIV, model an instruction cache and a data cache with a combined total cache space of 32KB (for a split cache, assume an 16KB instruction cache and an 16KB data cache. The block size should be varied (8B, 16B and 32B) and the associativity should be varied (direct-mapped, 2-way and 4-way). Model both split (separate instruction and data caches) and shared (all accesses go to a single cache that holds both instructions and data) caches. There is a total of 18 simulations. No other parameters should be varied. Graph the results you get from these experiments and discuss in detail why you see different trends in the graphs. Copy the trace file provided to the COE Linux system to run your study, but please gzip or compress the file when you are not using it.

Ans:

Given:

- The Total Cache Space = 32KB

  (i.e) for a split cache, assume 16KB for Instruction Space and 16KB for Data Cache

- The Block size should be varied as 8B, 16B and 32B
- The Associativity should be varied as Direct Map (1-way), 2-way and 4-way

Task:

The task is to model both split (separate instruction and data caches) and shared (all accesses go to a single cache that holds both instructions and data) caches

Split Cache Runs

1. **Block Size: 8B, Associativity: Direct Map**

   Command:

   dineroIV -l1-isize 16K -l1-ibsize 8 -l1-iassoc 1 -l1-dsize 16K -l1-dbsize 8 -l1-dassoc 1 -informat d < trace > out/split_b8_a1.txt

   Run Output:

```
---Simulation begins.
---Simulation complete.
l1-icache
 Metrics                      Total           Instrn           Data            Read            Write           Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches               559159          559159               0               0               0               0
  Fraction of total           1.0000          1.0000          0.0000          0.0000          0.0000          0.0000

 Demand Misses                 59975           59975               0               0               0               0
  Demand miss rate            0.1073          0.1073          0.0000          0.0000          0.0000          0.0000

 Multi-block refs                  0
 Bytes From Memory            479800
 ( / Demand Fetches)          0.8581
 Bytes To Memory                   0
 ( / Demand Writes)           0.0000
 Total Bytes r/w Mem          479800
 ( / Demand Fetches)          0.8581

l1-dcache
 Metrics                      Total           Instrn           Data            Read            Write           Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches               467428               0          467428          288238          179190               0
  Fraction of total           1.0000          0.0000          1.0000          0.6166          0.3834          0.0000

 Demand Misses                 42205               0           42205           25364           16841               0
  Demand miss rate            0.0903          0.0000          0.0903          0.0880          0.0940          0.0000

 Multi-block refs                  0
 Bytes From Memory            337640
 ( / Demand Fetches)          0.7223
 Bytes To Memory              198592
 ( / Demand Writes)           1.1083
 Total Bytes r/w Mem          536232
 ( / Demand Fetches)          1.1472

---Execution complete.
[sakthivelps@Eta out]$
```

2. **Block Size – 8B, Associativity: 2-way**

   Command:

   dineroIV -l1-isize 16K -l1-ibsize 8 -l1-iassoc 2 -l1-dsize 16K -l1-dbsize 8 -l1-dassoc 2 -informat d < trace > out/split_b8_a2.txt

   Run Output:

```
---Simulation begins.
---Simulation complete.
l1-icache
 Metrics                      Total           Instrn           Data            Read            Write           Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches               559159          559159               0               0               0               0
  Fraction of total           1.0000          1.0000          0.0000          0.0000          0.0000          0.0000

 Demand Misses                 47518           47518               0               0               0               0
  Demand miss rate            0.0850          0.0850          0.0000          0.0000          0.0000          0.0000

 Multi-block refs                  0
 Bytes From Memory            380144
 ( / Demand Fetches)          0.6798
 Bytes To Memory                   0
 ( / Demand Writes)           0.0000
 Total Bytes r/w Mem          380144
 ( / Demand Fetches)          0.6798

l1-dcache
 Metrics                      Total           Instrn           Data            Read            Write           Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches               467428               0          467428          288238          179190               0
  Fraction of total           1.0000          0.0000          1.0000          0.6166          0.3834          0.0000

 Demand Misses                 37299               0           37299           21731           15568               0
  Demand miss rate            0.0798          0.0000          0.0798          0.0754          0.0869          0.0000

 Multi-block refs                  0
 Bytes From Memory            298392
 ( / Demand Fetches)          0.6384
 Bytes To Memory              180688
 ( / Demand Writes)           1.0084
 Total Bytes r/w Mem          479080
 ( / Demand Fetches)          1.0249

---Execution complete.
[sakthivelps@Eta out]$
```

3. Block Size – 8B, Associativity: 4-way

Command:

dineroIV -l1-isize 16K -l1-ibsize 8 -l1-iassoc 4 -l1-dsize 16K -l1-dbsize 8 -l1-dassoc 4 -informat d < trace > out/split_b8_a4.txt

Run Output:

```
---Simulation begins.
---Simulation complete.
l1-icache
 Metrics                      Total           Instrn           Data            Read            Write           Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches               559159          559159               0               0               0               0
  Fraction of total           1.0000          1.0000          0.0000          0.0000          0.0000          0.0000

 Demand Misses                 44248           44248               0               0               0               0
  Demand miss rate            0.0791          0.0791          0.0000          0.0000          0.0000          0.0000

 Multi-block refs                 0
 Bytes From Memory            353984
 ( / Demand Fetches)          0.6331
 Bytes To Memory                   0
 ( / Demand Writes)           0.0000
 Total Bytes r/w Mem          353984
 ( / Demand Fetches)          0.6331

l1-dcache
 Metrics                      Total           Instrn           Data            Read            Write           Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches               467428               0          467428          288238          179190               0
  Fraction of total           1.0000          0.0000          1.0000          0.6166          0.3834          0.0000

 Demand Misses                 33991               0           33991           19717           14274               0
  Demand miss rate            0.0727          0.0000          0.0727          0.0684          0.0797          0.0000

 Multi-block refs                 0
 Bytes From Memory            271928
 ( / Demand Fetches)          0.5818
 Bytes To Memory              165552
 ( / Demand Writes)           0.9239
 Total Bytes r/w Mem          437480
 ( / Demand Fetches)          0.9359

---Execution complete.
[sakthivelps@Eta out]$
```

4. Block Size: 16B, Associativity: Direct Map

dineroIV -l1-isize 16K -l1-ibsize 16 -l1-iassoc 1 -l1-dsize 16K -l1-dbsize 16 -l1-dassoc 1 -informat d < trace > out/split_b16_a1.txt

```
---Simulation begins.
---Simulation complete.
l1-icache
 Metrics                      Total           Instrn          Data            Read            Write           Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches               559159          559159               0               0               0               0
  Fraction of total           1.0000          1.0000          0.0000          0.0000          0.0000          0.0000

 Demand Misses                 39402           39402               0               0               0               0
  Demand miss rate            0.0705          0.0705          0.0000          0.0000          0.0000          0.0000

 Multi-block refs                  0
 Bytes From Memory            630432
 ( / Demand Fetches)          1.1275
 Bytes To Memory                   0
 ( / Demand Writes)           0.0000
 Total Bytes r/w Mem          630432
 ( / Demand Fetches)          1.1275

l1-dcache
 Metrics                      Total           Instrn          Data            Read            Write           Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches               467428               0          467428          288238          179190               0
  Fraction of total           1.0000          0.0000          1.0000          0.6166          0.3834          0.0000

 Demand Misses                 34840               0           34840           22389           12451               0
  Demand miss rate            0.0745          0.0000          0.0745          0.0777          0.0695          0.0000

 Multi-block refs                  0
 Bytes From Memory            557440
 ( / Demand Fetches)          1.1926
 Bytes To Memory              321408
 ( / Demand Writes)           1.7937
 Total Bytes r/w Mem          878848
 ( / Demand Fetches)          1.8802

---Execution complete.
[sakthivelps@Eta out]$ |
```

5. **Block Size – 16B, Associativity: 2-way**

dineroIV -l1-isize 16K -l1-ibsize 16 -l1-iassoc 2 -l1-dsize 16K -l1-dbsize 16 -l1-dassoc 2 -informat d < trace > out/split_b16_a2.txt

```
---Simulation begins.
---Simulation complete.
l1-icache
 Metrics                      Total           Instrn          Data            Read            Write           Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches               559159          559159               0               0               0               0
  Fraction of total           1.0000          1.0000          0.0000          0.0000          0.0000          0.0000

 Demand Misses                 31062           31062               0               0               0               0
  Demand miss rate            0.0556          0.0556          0.0000          0.0000          0.0000          0.0000

 Multi-block refs                  0
 Bytes From Memory            496992
 ( / Demand Fetches)          0.8888
 Bytes To Memory                   0
 ( / Demand Writes)           0.0000
 Total Bytes r/w Mem          496992
 ( / Demand Fetches)          0.8888

l1-dcache
 Metrics                      Total           Instrn          Data            Read            Write           Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches               467428               0          467428          288238          179190               0
  Fraction of total           1.0000          0.0000          1.0000          0.6166          0.3834          0.0000

 Demand Misses                 30026               0           30026           18890           11136               0
  Demand miss rate            0.0642          0.0000          0.0642          0.0655          0.0621          0.0000

 Multi-block refs                  0
 Bytes From Memory            480416
 ( / Demand Fetches)          1.0278
 Bytes To Memory              285760
 ( / Demand Writes)           1.5947
 Total Bytes r/w Mem          766176
 ( / Demand Fetches)          1.6391

---Execution complete.
[sakthivelps@Eta out]$ |
```

6. **Block Size – 16B, Associativity: 4-way**

dineroIV -l1-isize 16K -l1-ibsize 16 -l1-iassoc 4 -l1-dsize 16K -l1-dbsize 16 -l1-dassoc 4 -informat d < trace > out/split_b16_a4.txt

```
---Simulation begins.
---Simulation complete.
l1-icache
 Metrics                      Total           Instrn          Data            Read            Write           Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches               559159          559159               0               0               0               0
  Fraction of total           1.0000          1.0000          0.0000          0.0000          0.0000          0.0000

 Demand Misses                 29015           29015               0               0               0               0
  Demand miss rate            0.0519          0.0519          0.0000          0.0000          0.0000          0.0000

 Multi-block refs                  0
 Bytes From Memory            464240
 ( / Demand Fetches)          0.8302
 Bytes To Memory                   0
 ( / Demand Writes)           0.0000
 Total Bytes r/w Mem          464240
 ( / Demand Fetches)          0.8302

l1-dcache
 Metrics                      Total           Instrn          Data            Read            Write           Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches               467428               0          467428          288238          179190               0
  Fraction of total           1.0000          0.0000          1.0000          0.6166          0.3834          0.0000

 Demand Misses                 26557               0           26557           16304           10253               0
  Demand miss rate            0.0568          0.0000          0.0568          0.0566          0.0572          0.0000

 Multi-block refs                  0
 Bytes From Memory            424912
 ( / Demand Fetches)          0.9090
 Bytes To Memory              255168
 ( / Demand Writes)           1.4240
 Total Bytes r/w Mem          680080
 ( / Demand Fetches)          1.4549

---Execution complete.
[sakthivelps@Eta out]$
```

7. **Block Size: 32B, Associativity: Direct Map**

dineroIV -l1-isize 16K -l1-ibsize 32 -l1-iassoc 1 -l1-dsize 16K -l1-dbsize 32 -l1-dassoc 1 -informat d < trace > out/split_b32_a1.txt

```
---Simulation begins.
---Simulation complete.
l1-icache
 Metrics                      Total           Instrn          Data            Read            Write           Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches               559159          559159               0               0               0               0
  Fraction of total           1.0000          1.0000          0.0000          0.0000          0.0000          0.0000

 Demand Misses                 26578           26578               0               0               0               0
  Demand miss rate            0.0475          0.0475          0.0000          0.0000          0.0000          0.0000

 Multi-block refs                  0
 Bytes From Memory            850496
 ( / Demand Fetches)          1.5210
 Bytes To Memory                   0
 ( / Demand Writes)           0.0000
 Total Bytes r/w Mem          850496
 ( / Demand Fetches)          1.5210

l1-dcache
 Metrics                      Total           Instrn          Data            Read            Write           Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches               467428               0          467428          288238          179190               0
  Fraction of total           1.0000          0.0000          1.0000          0.6166          0.3834          0.0000

 Demand Misses                 32649               0           32649           22446           10203               0
  Demand miss rate            0.0698          0.0000          0.0698          0.0779          0.0569          0.0000

 Multi-block refs                  0
 Bytes From Memory           1044768
 ( / Demand Fetches)          2.2351
 Bytes To Memory              592256
 ( / Demand Writes)           3.3052
 Total Bytes r/w Mem         1637024
 ( / Demand Fetches)          3.5022

---Execution complete.
[sakthivelps@Eta out]$
```

8. **Block Size – 32B, Associativity: 2-way**

dineroIV -l1-isize 16K -l1-ibsize 32 -l1-iassoc 2 -l1-dsize 16K -l1-dbsize 32 -l1-dassoc 2 -informat d < trace > out/split_b32_a2.txt

```
---Simulation begins.
---Simulation complete.
l1-icache
 Metrics                     Total           Instrn          Data            Read            Write           Misc
 -----------------           ------          ------          ------          ------          ------          ------
 Demand Fetches              559159          559159               0               0               0               0
  Fraction of total          1.0000          1.0000          0.0000          0.0000          0.0000          0.0000

 Demand Misses                21152           21152               0               0               0               0
  Demand miss rate           0.0378          0.0378          0.0000          0.0000          0.0000          0.0000

 Multi-block refs                 0
 Bytes From Memory           676864
 ( / Demand Fetches)         1.2105
 Bytes To Memory                  0
 ( / Demand Writes)          0.0000
 Total Bytes r/w Mem         676864
 ( / Demand Fetches)         1.2105

l1-dcache
 Metrics                     Total           Instrn          Data            Read            Write           Misc
 -----------------           ------          ------          ------          ------          ------          ------
 Demand Fetches              467428               0          467428          288238          179190               0
  Fraction of total          1.0000          0.0000          1.0000          0.6166          0.3834          0.0000

 Demand Misses                27606               0           27606           18533            9073               0
  Demand miss rate           0.0591          0.0000          0.0591          0.0643          0.0506          0.0000

 Multi-block refs                 0
 Bytes From Memory           883392
 ( / Demand Fetches)         1.8899
 Bytes To Memory             513856
 ( / Demand Writes)          2.8677
 Total Bytes r/w Mem        1397248
 ( / Demand Fetches)         2.9892

---Execution complete.
[sakthivelps@Eta out]$ |
```

9. **Block Size – 32B, Associativity: 4-way**

dineroIV -l1-isize 16K -l1-ibsize 32 -l1-iassoc 4 -l1-dsize 16K -l1-dbsize 32 -l1-dassoc 4 -informat d < trace > out/split_b32_a4.txt

```
---Simulation begins.
---Simulation complete.
l1-icache
 Metrics                     Total           Instrn          Data            Read            Write           Misc
 -----------------           ------          ------          ------          ------          ------          ------
 Demand Fetches              559159          559159               0               0               0               0
  Fraction of total          1.0000          1.0000          0.0000          0.0000          0.0000          0.0000

 Demand Misses                20148           20148               0               0               0               0
  Demand miss rate           0.0360          0.0360          0.0000          0.0000          0.0000          0.0000

 Multi-block refs                 0
 Bytes From Memory           644736
 ( / Demand Fetches)         1.1530
 Bytes To Memory                  0
 ( / Demand Writes)          0.0000
 Total Bytes r/w Mem         644736
 ( / Demand Fetches)         1.1530

l1-dcache
 Metrics                     Total           Instrn          Data            Read            Write           Misc
 -----------------           ------          ------          ------          ------          ------          ------
 Demand Fetches              467428               0          467428          288238          179190               0
  Fraction of total          1.0000          0.0000          1.0000          0.6166          0.3834          0.0000

 Demand Misses                24071               0           24071           15729            8342               0
  Demand miss rate           0.0515          0.0000          0.0515          0.0546          0.0466          0.0000

 Multi-block refs                 0
 Bytes From Memory           770272
 ( / Demand Fetches)         1.6479
 Bytes To Memory             451168
 ( / Demand Writes)          2.5178
 Total Bytes r/w Mem        1221440
 ( / Demand Fetches)         2.6131

---Execution complete.
[sakthivelps@Eta out]$ |
```

Shared Cache Runs

10. **Block Size: 8B, Associativity: Direct Map**

Command:

dineroIV -l1-usize 32K -l1-ubsize 8 -l1-uassoc 1 -informat d < trace > out/shared_b8_a1.txt

```
---Summary of options (-help option gives usage information).

-l1-usize 32768
-l1-ubsize 8
-l1-usbsize 8
-l1-uassoc 1
-l1-urepl l
-l1-ufetch d
-l1-uwalloc a
-l1-uwback a
-skipcount 0
-flushcount 0
-maxcount 0
-stat-interval 0
-informat d
-on-trigger 0x0
-off-trigger 0x0

---Simulation begins.
---Simulation complete.
l1-ucache
 Metrics                      Total           Instrn           Data            Read           Write            Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches              1026587          559159          467428          288238          179190               0
  Fraction of total           1.0000          0.5447          0.4553          0.2808          0.1745          0.0000

 Demand Misses                 97518           52628           44890           28300           16590               0
  Demand miss rate            0.0950          0.0941          0.0960          0.0982          0.0926          0.0000

 Multi-block refs                 0
 Bytes From Memory            780144
 ( / Demand Fetches)          0.7599
 Bytes To Memory              214880
 ( / Demand Writes)           1.1992
 Total Bytes r/w Mem          995024
 ( / Demand Fetches)          0.9693

---Execution complete.
[sakthivelps@Eta out]$
```

11. Block Size: 8B, Associativity: 2-way

dineroIV -l1-usize 32K -l1-ubsize 8 -l1-uassoc 2 -informat d < trace > out/shared_b8_a2.txt

```
---Summary of options (-help option gives usage information).

-l1-usize 32768
-l1-ubsize 8
-l1-usbsize 8
-l1-uassoc 2
-l1-urepl l
-l1-ufetch d
-l1-uwalloc a
-l1-uwback a
-skipcount 0
-flushcount 0
-maxcount 0
-stat-interval 0
-informat d
-on-trigger 0x0
-off-trigger 0x0

---Simulation begins.
---Simulation complete.
l1-ucache
 Metrics                      Total           Instrn           Data            Read           Write            Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches              1026587          559159          467428          288238          179190               0
  Fraction of total           1.0000          0.5447          0.4553          0.2808          0.1745          0.0000

 Demand Misses                 76564           40993           35571           21737           13834               0
  Demand miss rate            0.0746          0.0733          0.0761          0.0754          0.0772          0.0000

 Multi-block refs                 0
 Bytes From Memory            612512
 ( / Demand Fetches)          0.5966
 Bytes To Memory              173424
 ( / Demand Writes)           0.9678
 Total Bytes r/w Mem          785936
 ( / Demand Fetches)          0.7656

---Execution complete.
[sakthivelps@Eta out]$
```

12. Block Size: 8B, Associativity: 4-way

dineroIV -l1-usize 32K -l1-ubsize 8 -l1-uassoc 4 -informat d < trace > out/shared_b8_a4.txt

```
---Summary of options (-help option gives usage information).

-l1-usize 32768
-l1-ubsize 8
-l1-usbsize 8
-l1-uassoc 4
-l1-urepl l
-l1-ufetch d
-l1-uwalloc a
-l1-uwback a
-skipcount 0
-flushcount 0
-maxcount 0
-stat-interval 0
-informat d
-on-trigger 0x0
-off-trigger 0x0

---Simulation begins.
---Simulation complete.
l1-ucache
 Metrics                      Total           Instrn           Data           Read          Write           Misc
 -----------------           ------           ------         ------         ------         ------         ------
 Demand Fetches             1026587           559159         467428         288238         179190              0
  Fraction of total          1.0000           0.5447         0.4553         0.2808         0.1745         0.0000

 Demand Misses                70927            37361          33566          20065          13501              0
  Demand miss rate           0.0691           0.0668         0.0718         0.0696         0.0753         0.0000

 Multi-block refs                 0
 Bytes From Memory           567416
 ( / Demand Fetches)         0.5527
 Bytes To Memory             168456
 ( / Demand Writes)          0.9401
 Total Bytes r/w Mem         735872
 ( / Demand Fetches)         0.7168

---Execution complete.
[sakthivelps@Eta out]$
```

13. Block Size: 16B, Associativity: Direct Map

Command:

dineroIV -l1-usize 32K -l1-ubsize 16 -l1-uassoc 1 -informat d < trace > out/shared_b16_a1.txt

```
---Summary of options (-help option gives usage information).

-l1-usize 32768
-l1-ubsize 16
-l1-usbsize 16
-l1-uassoc 1
-l1-urepl l
-l1-ufetch d
-l1-uwalloc a
-l1-uwback a
-skipcount 0
-flushcount 0
-maxcount 0
-stat-interval 0
-informat d
-on-trigger 0x0
-off-trigger 0x0

---Simulation begins.
---Simulation complete.
l1-ucache
 Metrics                      Total           Instrn           Data           Read          Write           Misc
 -----------------           ------           ------         ------         ------         ------         ------
 Demand Fetches             1026587           559159         467428         288238         179190              0
  Fraction of total          1.0000           0.5447         0.4553         0.2808         0.1745         0.0000

 Demand Misses                71219            36185          35034          23673          11361              0
  Demand miss rate           0.0694           0.0647         0.0750         0.0821         0.0634         0.0000

 Multi-block refs                 0
 Bytes From Memory          1139504
 ( / Demand Fetches)         1.1100
 Bytes To Memory             338880
 ( / Demand Writes)          1.8912
 Total Bytes r/w Mem        1478384
 ( / Demand Fetches)         1.4401

---Execution complete.
[sakthivelps@Eta out]$
```

14. Block Size: 16B, Associativity: 2-way

dineroIV -l1-usize 32K -l1-ubsize 16 -l1-uassoc 2 -informat d < trace > out/shared_b16_a2.txt

```
---Summary of options (-help option gives usage information).

-l1-usize 32768
-l1-ubsize 16
-l1-usbsize 16
-l1-uassoc 2
-l1-urepl l
-l1-ufetch d
-l1-uwalloc a
-l1-uwback a
-skipcount 0
-flushcount 0
-maxcount 0
-stat-interval 0
-informat d
-on-trigger 0x0
-off-trigger 0x0

---Simulation begins.
---Simulation complete.
l1-ucache
 Metrics                      Total           Instrn          Data            Read            Write           Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches              1026587          559159          467428          288238          179190               0
  Fraction of total           1.0000          0.5447          0.4553          0.2808          0.1745          0.0000

 Demand Misses                 55801           28492           27309           18043            9266               0
  Demand miss rate            0.0544          0.0510          0.0584          0.0626          0.0517          0.0000

 Multi-block refs                 0
 Bytes From Memory            892816
 ( / Demand Fetches)          0.8697
 Bytes To Memory              268208
 ( / Demand Writes)           1.4968
 Total Bytes r/w Mem         1161024
 ( / Demand Fetches)          1.1310

---Execution complete.
[sakthivelps@Eta out]$ 
```

15. Block Size: 16B, Associativity: 4-way

dineroIV -l1-usize 32K -l1-ubsize 16 -l1-uassoc 4 -informat d < trace > out/shared_b16_a4.txt

```
---Summary of options (-help option gives usage information).

-l1-usize 32768
-l1-ubsize 16
-l1-usbsize 16
-l1-uassoc 4
-l1-urepl l
-l1-ufetch d
-l1-uwalloc a
-l1-uwback a
-skipcount 0
-flushcount 0
-maxcount 0
-stat-interval 0
-informat d
-on-trigger 0x0
-off-trigger 0x0

---Simulation begins.
---Simulation complete.
l1-ucache
 Metrics                      Total           Instrn          Data            Read            Write           Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches              1026587          559159          467428          288238          179190               0
  Fraction of total           1.0000          0.5447          0.4553          0.2808          0.1745          0.0000

 Demand Misses                 50337           25411           24926           15931            8995               0
  Demand miss rate            0.0490          0.0454          0.0533          0.0553          0.0502          0.0000

 Multi-block refs                 0
 Bytes From Memory            805392
 ( / Demand Fetches)          0.7845
 Bytes To Memory              253824
 ( / Demand Writes)           1.4165
 Total Bytes r/w Mem         1059216
 ( / Demand Fetches)          1.0318

---Execution complete.
[sakthivelps@Eta out]$ 
```

16. Block Size: 32B, Associativity: Direct Map

Command:

dineroIV -l1-usize 32K -l1-ubsize 32 -l1-uassoc 1 -informat d < trace > out/shared_b32_a1.txt

```
---Summary of options (-help option gives usage information).

-l1-usize 32768
-l1-ubsize 32
-l1-usbsize 32
-l1-uassoc 1
-l1-urepl l
-l1-ufetch d
-l1-uwalloc a
-l1-uwback a
-skipcount 0
-flushcount 0
-maxcount 0
-stat-interval 0
-informat d
-on-trigger 0x0
-off-trigger 0x0

---Simulation begins.
---Simulation complete.
l1-ucache
 Metrics                      Total           Instrn          Data            Read            Write           Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches               1026587         559159          467428          288238          179190               0
  Fraction of total           1.0000          0.5447          0.4553          0.2808          0.1745          0.0000

 Demand Misses                  56913           26459           30454           22060            8394               0
  Demand miss rate            0.0554          0.0473          0.0652          0.0765          0.0468          0.0000

 Multi-block refs                   0
 Bytes From Memory            1821216
 ( / Demand Fetches)           1.7740
 Bytes To Memory               598016
 ( / Demand Writes)            3.3373
 Total Bytes r/w Mem          2419232
 ( / Demand Fetches)           2.3566

---Execution complete.
[sakthivelps@Eta out]$
```

17. Block Size: 32B, Associativity: 2-way

dineroIV -l1-usize 32K -l1-ubsize 32 -l1-uassoc 2 -informat d < trace > out/shared_b32_a2.txt

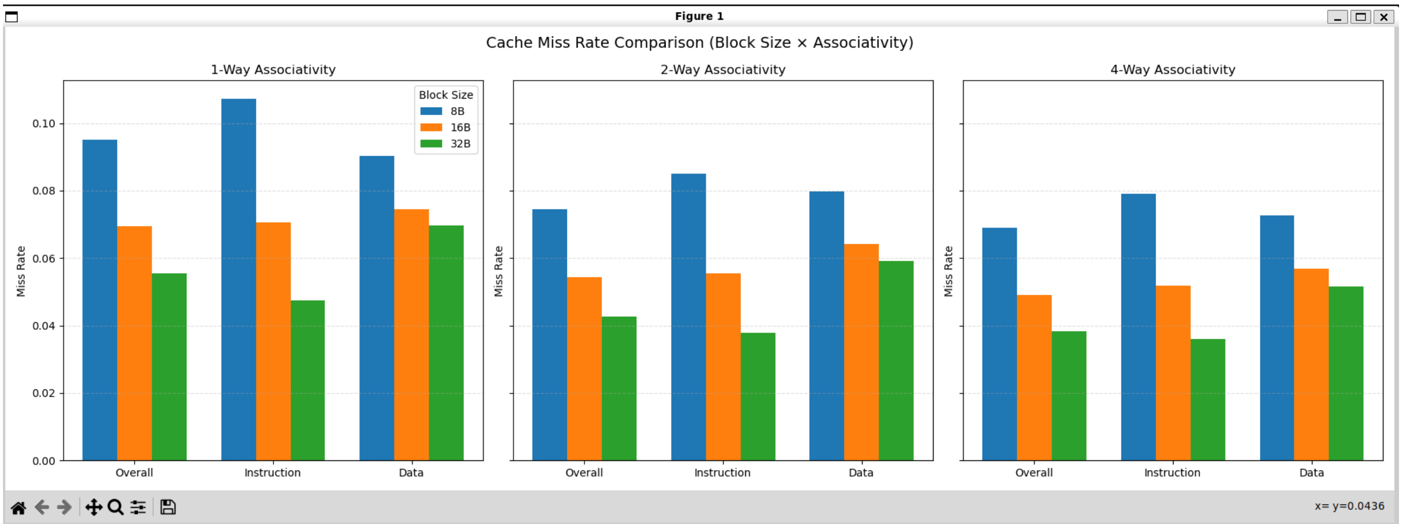
```
---Summary of options (-help option gives usage information).

-l1-usize 32768
-l1-ubsize 32
-l1-usbsize 32
-l1-uassoc 2
-l1-urepl l
-l1-ufetch d
-l1-uwalloc a
-l1-uwback a
-skipcount 0
-flushcount 0
-maxcount 0
-stat-interval 0
-informat d
-on-trigger 0x0
-off-trigger 0x0

---Simulation begins.
---Simulation complete.
l1-ucache
 Metrics                      Total           Instrn          Data            Read            Write           Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches               1026587         559159          467428          288238          179190               0
  Fraction of total           1.0000          0.5447          0.4553          0.2808          0.1745          0.0000

 Demand Misses                  43853           20857           22996           16324            6672               0
  Demand miss rate            0.0427          0.0373          0.0492          0.0566          0.0372          0.0000

 Multi-block refs                   0
 Bytes From Memory            1403296
 ( / Demand Fetches)           1.3670
 Bytes To Memory               459904
 ( / Demand Writes)            2.5666
 Total Bytes r/w Mem          1863200
 ( / Demand Fetches)           1.8149

---Execution complete.
[sakthivelps@Eta out]$
```

18. Block Size: 32B, Associativity: 4-way

dineroIV -l1-usize 32K -l1-ubsize 32 -l1-uassoc 4 -informat d < trace > out/shared_b32_a4.txt

```
---Summary of options (-help option gives usage information).

-l1-usize 32768
-l1-ubsize 32
-l1-usbsize 32
-l1-uassoc 4
-l1-urepl l
-l1-ufetch d
-l1-uwalloc a
-l1-uwback a
-skipcount 0
-flushcount 0
-maxcount 0
-stat-interval 0
-informat d
-on-trigger 0x0
-off-trigger 0x0

---Simulation begins.
---Simulation complete.
l1-ucache
 Metrics                      Total           Instrn           Data            Read           Write            Misc
 -----------------            ------          ------          ------          ------          ------          ------
 Demand Fetches             1026587          559159          467428          288238          179190               0
  Fraction of total          1.0000          0.5447          0.4553          0.2808          0.1745          0.0000

 Demand Misses                39377           18650           20727           14247            6480               0
  Demand miss rate           0.0384          0.0334          0.0443          0.0494          0.0362          0.0000

 Multi-block refs                 0
 Bytes From Memory          1260064
 ( / Demand Fetches)         1.2274
 Bytes To Memory             429728
 ( / Demand Writes)          2.3982
 Total Bytes r/w Mem        1689792
 ( / Demand Fetches)         1.6460

---Execution complete.
[sakthivelps@Eta out]$ |
```

Overall Cache Miss Rate Comparison Plot



Effect of Block Size:

- Larger block sizes consistently reduce miss rates across all associativity levels.
- 8B blocks fetch too little useful data whereas, 16B and 32B blocks capture more spatial locality (fewer misses).
- Improvement from 16B to 32B is smaller due to diminishing returns.

Effect of Associativity:

- Higher associativity reduces conflict misses.
- Direct-mapped caches show the worst miss rates because every index allows only one line.
- 2-way associativity eliminates most conflicts thus leading to large drop-in miss rate.
- 4-way provides additional reduction but with smaller gains.

Split vs Shared Caches:

- Split caches generally perform better than a unified 32KB cache.
- Shared cache forces instruction & data lines to compete thus more conflict and capacity pressure.
- Split structure isolates streams therefore the data accesses cannot evict instructions and vice-versa. Reduced interference leads to lower miss rates, especially for instruction fetches.

Instruction vs Data Miss Rates:

- Instruction miss rates are consistently lower across all configurations and the access pattern is mostly sequential (strong spatial locality).
- Data accesses are more irregular (stack operations, pointer chasing, offsets).
- Data miss rates are more sensitive to associativity and block size.